



Effects of Early Antifungal Prophylaxis Regimen Combinations on Survival Following Lung Transplantation

Scott Cohen, Kathryn DeSear, Mattia Prosperi, Moustafa Younis, Cynthia Gries

College of Medicine, College of Pharmacy, and College of Public Health and Health Professions

INTRODUCTION

BACKGROUND

- Invasive fungal infections are a leading cause of early mortality after lung transplantation.
- Early initiation of antifungal agents may reduce infection related mortality, yet prophylaxis strategies vary widely across centers, and the benefits of specific classes and multidrug combinations, including micafungin bridging, remain uncertain.
- Clarifying regimen specific survival benefits is essential for standardizing and optimizing post transplant prophylaxis.

OBJECTIVE:

- To evaluate the impact of early antifungal prophylaxis on post lung transplant survival, and to characterize heterogeneity of effect across antifungal classes and combination strategies.

METHODS

STUDY DESIGN

- Retrospective cohort of incident lung transplant cases from 2010 to 2024 at a tertiary-care hospital
- Data Source:** Clinical, laboratory, microbiology, flowsheet data extracted from the institution's clinical data warehouse
- Exposures:** Early antifungal prophylaxis (<48 hours after transplant)
Antifungal Classes: any azole, micafungin, inhaled amphotericin, class combinations, or no antifungal prophylaxis.
- Outcome:** All-cause mortality at 180 days and 1 year
- Covariates:** Demographics, clinical status at transplant (ICU, comorbidity), prior infection history, transplant year

ANALYSIS

- Cohort summarized descriptively (Table 1, Figure 1).
- Marginal structural Cox models to estimate regimen effects (Figure 2 and 3) [See "Exposure Model Details"]
- Subgroup analyses for interactions (Wald) between antifungal classes and combination strategies (Figure 3).

EXPOSURE MODEL DETAILS

- Stabilized weights were generated from models including all covariates, using generalized boosted models for binary exposures and multinomial logistic regression for regimen comparisons.

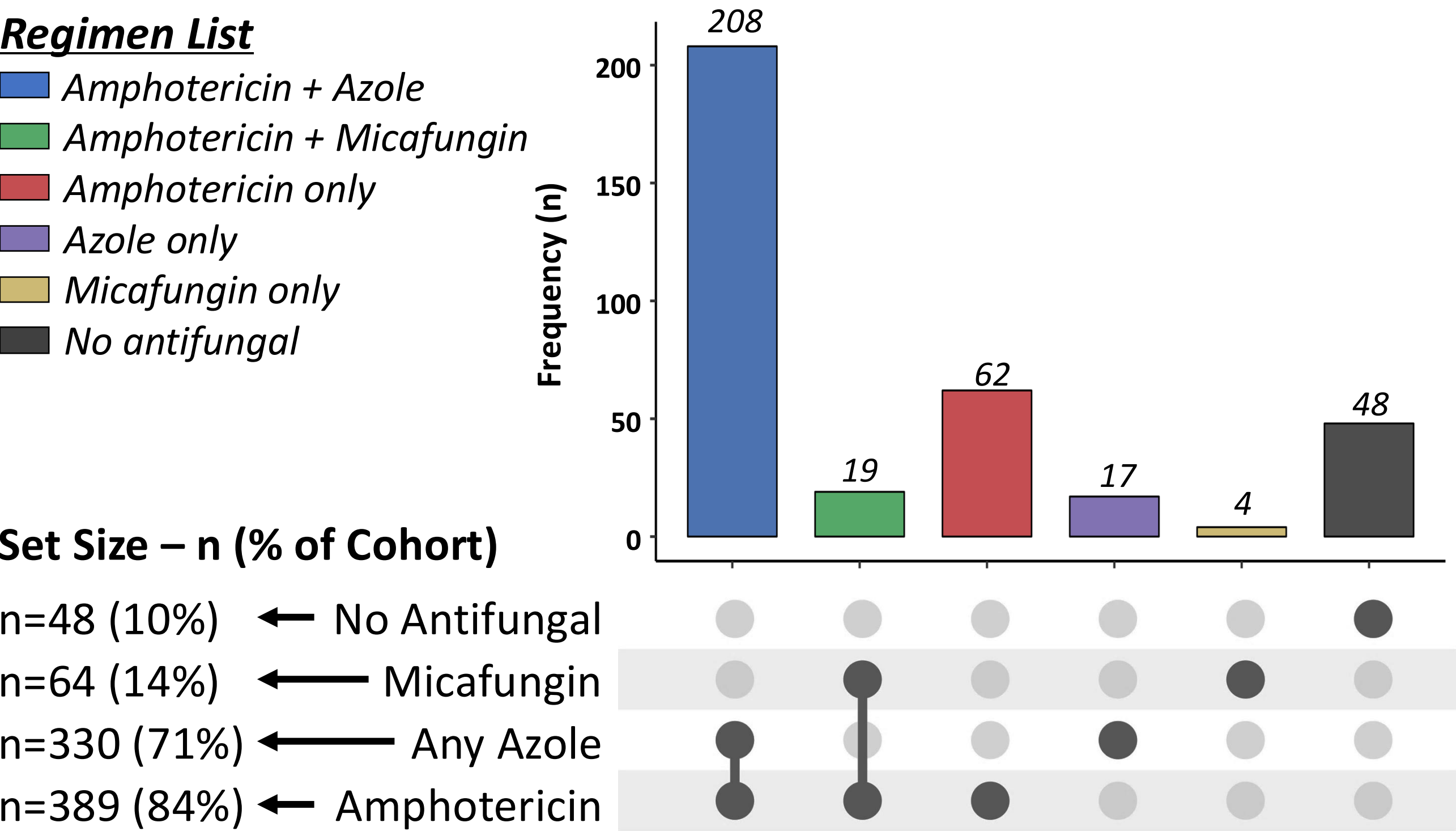
Diagnostic	Description/Results
Maximum SMD after weighting	<0.10 to 0.15 across all models
Covariates meeting SMD < 0.10	>95% on all models, majority 100%
Extreme weights	Stabilization, trimming (99 th Percent)

RESULTS OVERVIEW: TABLES & FIGURES

Table 1: Descriptive Summary of Study Cohort

Characteristic	Overall (n = 463)
Follow-Up (Years) - Median	5.3
Age at Transplant - Years (SD)	56.0 (14.3)
Sex (Female) - n (%)	176 (38%)
Race (White) - n (%)	376 (81.2%)
Ethnicity (Hispanic) - n (%)	71 (15.3%)
Transplant Type - n (%)	
Double Lung	395 (85.3%)
Single Lung	68 (14.7%)
Year of Transplant - n (%)	
2010-2014	65 (14%)
2015-2019	184 (39.7%)
2020+	214 (46.2%)
Prior Comorbidities – n (%)	
Chronic Pulmonary Disease	416 (89.8%)
Peripheral Vascular Disease	287 (62%)
Congestive Heart Failure	190 (41%)
Cerebrovascular Disease	121 (26.1%)
Diabetes w/o complications	109 (23.5%)
Renal disease	103 (22.2%)
Mild Liver Disease	93 (20.1%)
Overall Survival (Adjusted) – %	
180-Day Survival	95.2%
1-Year Survival	92.5%
5-Year Survival	66.8%

Figure 1: UpSet Plot Illustrating the Distribution of Early Antifungal Prophylaxis Regimens Among Study Cohort



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Contributing Authors: Sai Chintalapati, Victoria Reams, Gabrielle Bradley, Amir Emtiazjoo, Mindaugas Rackauskas, Mohammad Aladaileh, Brandon Janssen, Adriana Zuniga-Valverde

Figure 2: Adjusted Effects of Early Antifungals on 180 Day and 1 Year Survival

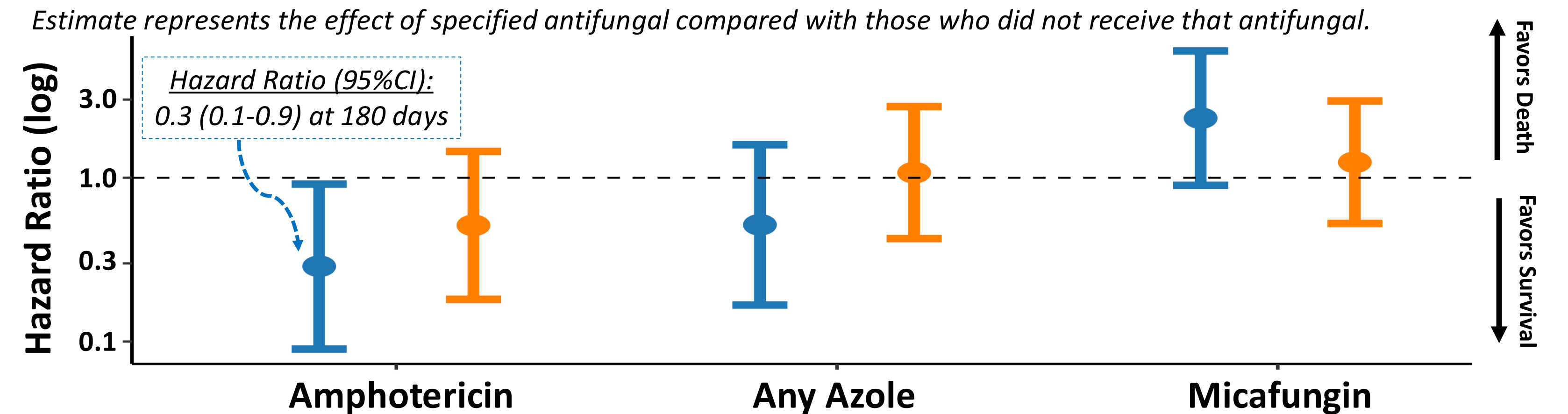


Table 2: Adjusted Treatment Effects of Early Amphotericin-Micafungin and Amphotericin-Azole Combination Regimens on Post-Transplant Survival

Estimates represent the effect of each regimen compared with recipients who did not receive any antifungal in that combination.

	Time Horizon	Regimen (Ref=None)	Hazard Ratio (95%CI)	P Value	P for Interaction
Amphotericin Micafungin	180 Days	Ampho Only	0.55 (0.16, 1.89)	0.340	0.601
		Micafungin Only	5.84 (1.06, 32.3)	0.043	
		Ampho + Micafungin	1.80 (0.51, 6.40)	0.361	
	1 Year	Ampho Only	1.27 (0.50, 3.26)	0.614	0.178
		Micafungin Only	4.91 (0.88, 27.4)	0.070	
		Ampho + Micafungin	1.53 (0.44, 5.32)	0.506	
Amphotericin Any Azole	180 Days	Ampho Only	0.09 (0.01, 0.85)	0.036	0.198
		Azole Only	0.42 (0.04, 4.37)	0.469	
		Ampho + Azole	0.29 (0.08, 1.06)	0.061	
	1 Year	Ampho Only	0.29 (0.04, 1.94)	0.202	0.446
		Azole Only	0.73 (0.12, 4.50)	0.737	
		Ampho + Azole	0.54 (0.15, 1.92)	0.340	

CONCLUSION

- Early initiation of inhaled amphotericin after lung transplantation was associated with improved survival.
- Combination regimens that included micafungin did not confer additional survival benefit
- Several Limitations: Single-center, retrospective, limited adjustment for confounding (time-varying), sparse power for combinations, and absence of secondary outcomes
- Future prospective studies integrating fungal isolate data and antifungal resistance patterns are needed to further refine prophylactic strategies.